Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} 0 < \kappa_\nu(u) &= \det(f_u(u,0), f_{vv}(u,0), f_{uu}(u,0)) \\ &= \det(T\circ \tilde f_u(u,0), T\circ \tilde f_{vv}(u,0), T\circ \tilde f_{uu}(u,0)) \\ &= \det(\tilde f_u(u,0), \tilde f_{vv}(u,0), \tilde f_{uu}(u,0)) \\ &= \xi_u(u,0)\det(\check f_\xi(\xi(u,0),0), \check f_{\eta\eta}(\xi(u,0),0), \check f_{\xi\xi}(\xi(u,0),0)). \end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \mathcal{Q}_t(x_{1:t-1}) = \rho_t\Big(\mathfrak{Q}_t(x_{1:t-1},\xi_t)\Big) \quad &\geq \quad \rho_t\Big(\mathfrak{Q}_t^{k-1}(x_{1:t-1},\xi_t)\Big) \\ &\geq \quad \sup_{p\in\mathcal{P}_t}\sum_{j=1}^M p_j \Phi_{t,j}\mathfrak{Q}_t^{k-1}(x_{1:t-1},\xi_{t,j}). \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\left[\begin{array}{cccc} g_2+{w_2}^2h_4+{n_2}^2h_5 & w_2w_3h_4+n_2n_3h_5 & w_2h_4 & n_2h_5 \\ w_2w_3h_4+n_2n_3h_5 & g_3+{w_3}^2h_4+{n_3}^2h_5 & w_3h_4 & n_3h_5 \\ w_2h_4 & w_3h_4 & h_4 & 0 \\ n_2h_5 & n_3h_5 & 0 & h_5 \end{array}\right],$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{array}{ll} D^{++} = u^{+\alpha}\frac{\partial}{\partial u^{-\alpha}}; & D^{--} = u^{-\alpha}\frac{\partial}{\partial u^{+\alpha}} \\ 2D^{++} = [D^0, D^{++}]; & -2D^{--} = [D^0, D^{--}] \\ D^0 = [D^{++}, D^{--}] = u^{+\alpha}\frac{\partial}{\partial u^{+\alpha}} - u^{-\alpha}\frac{\partial}{\partial u^{-\alpha}} & \end{array}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\zeta_\nu(s) = -\frac{1}{\pi}\Re\left\{x^{1-s}\,e^{-\imath\frac{\pi}{2}s}\int_0^\infty (y-\imath)^{-s}\,\frac{\Delta_\nu^{TE'}(\imath x(y-\imath))}{\Delta_\nu^{TE}(\imath x(y-\imath))}\,dy\right\},$$